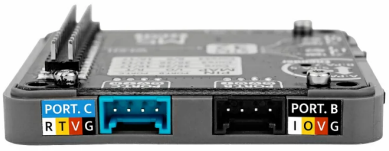
Module HMI

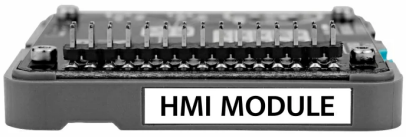
SKU:M129



















Description

Module HMI is a human-machine interaction module that provides a **rotary encoder**, **two input buttons**, and **two LED indicators**. It uses the STM32F030 as the acquisition and communication MCU, communicating with the M5 host via I2C. Additionally, the module features **PORT.B** and **PORT.C** interfaces and includes a built-in **500mAh** lithium battery. This module is suitable for various applications requiring handheld interactive operations.

Features

- STM32F030F4P6, ARM Cortex-M0 @ 16 KB Flash and 4 KB SRAM
- Human-machine interaction (rotary encoder, two input buttons, and two LED indicators)
- PORT B and PORT C interfaces
- Programming platforms: Arduino, UIFlow

Includes

- 1× Module HMI

Applications

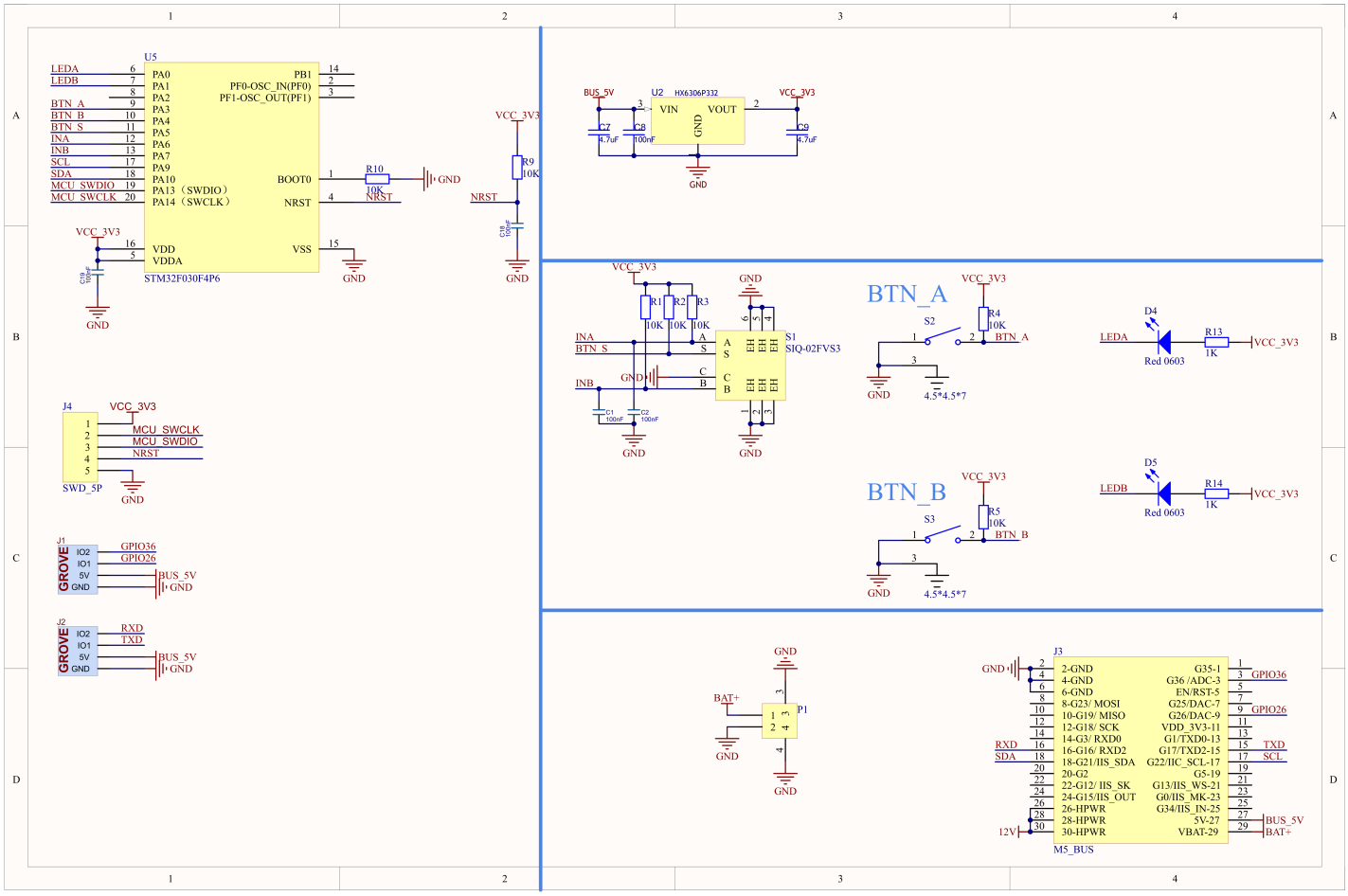
- Industrial control
- Embedded systems
- Smart home

Specifications

Specification	Parameter
MCU	STM32F030F4P6
Communication	I2C @0x41
Interactive Peripherals	Rotary Encoder, two input buttons, and two indicator LEDs
Battery	500mAh Li-polymer battery
Input Voltage	5V
Operating Temperature	0°C ~ 40°C
Product Size	54.0 x 55.8 x 13.0mm
Product Weight	22.8g
Package Size	95.0 x 77.0 x 14.0mm
Gross Weight	30.0g

Schematics

- Module HMI Schematics PDF



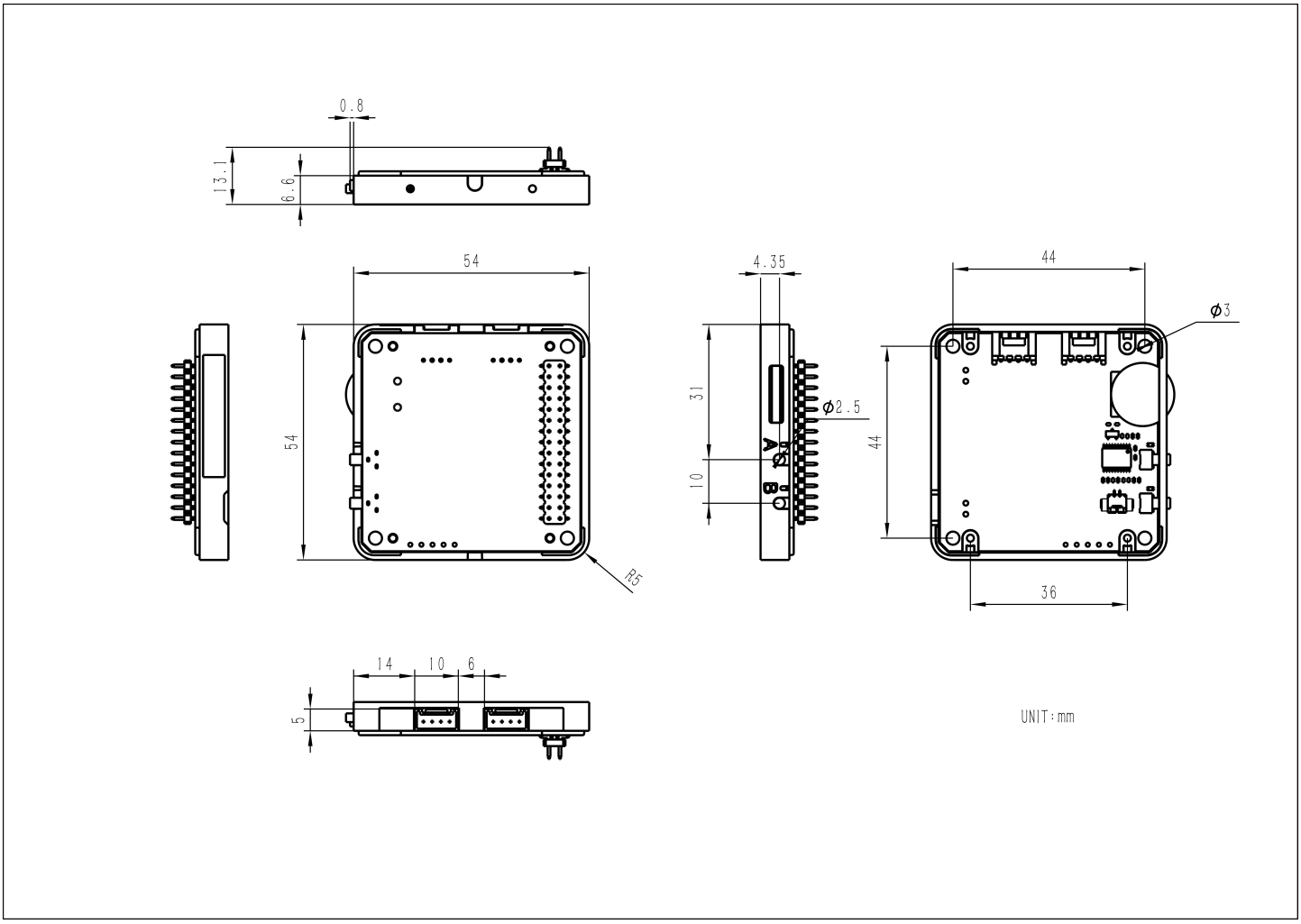
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	PORT.B
GND	5	6	
	7	8	PORT.B
	9	10	
	11	12	
	13	14	
TXD	15	16	RXD
SDA	17	18	SCL
	19	20	
	21	22	
	23	24	
HPWR	25	26	
HPWR	27	28	5V
HPWR	29	30	

Model Size

Module HMI Model Size PDF



Datasheets

- STM32F030F4P6 Datasheet

Softwares

Arduino

- Module HMI Arduino Library

UiFlow1

- Module HMI UiFlow1 Docs

UiFlow2

- Module HMI UiFlow2 Docs

Internal Firmware

- Module HMI Internal Firmware

Protocol

M5Stack Module HMI I2C Protocol

V1 (FW Version)
2022/11/11

REG MAP (Addr:0x41)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Counter Value (accumulate)	0x00 R/W	Cnt-byte0	Cnt-byte1	Cnt-byte2	Cnt-byte3													**Cnt:** -2147483648-2147483647 (will be reset after set reg 0x40)
Increment Value	0x10 R	Inc-byte0	Inc-byte1	Inc-byte2	Inc-byte3													**Cnt:** -2147483648-2147483647 (will be reset after get)
Button Value	0x20 R	Btn-S	Btn-1	Btn-2														**BNT:** 0~1
LED	0x30 R/W	LED-1	LED-2															**LED:** 1, LED on; 0, LED off
Counter Reset	0x40 W	Cnt-RST																**RST:** write 1 to reset counter
Firmware Version	0xF0 R															Version		**Version:** firmware version number
I2C Address	0xF0 R/W																Address	**Address:** 1~127

Video